

Intelligent Portfolio Optimization Dashboard

An Autonomous Hybrid Framework Combining Classical Modern Portfolio Theory (MPT) and Deep Reinforcement Learning (PPO)

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DEPLOYMENT

Streamlit Cloud

1. Executive Summary

This project presents an autonomous robo-advisory engine for dynamic asset allocation across traditional equities and digital assets (AAPL, MSFT, GOOGL, AMZN, BTC-USD). It integrates classical **Markowitz Mean-Variance Optimization** with continuous **Deep Reinforcement Learning (PPO)**, resolving cross-market latency and structural regime shifts.

MPT MAX SHARPE	ADAPTIVE RL (PPO)	EQUAL WEIGHT INDEX
\$18,876.47	\$11,656.81	\$11,846.22
+88.8% Total Return	+16.6% Dynamic Return	+18.5% Benchmark Return

2. System Architecture & Cross-Market Data Synchronization

A forward-fill (ffill()) pipeline synchronizes 5-day stock exchanges with 24/7 digital cryptocurrency timelines without introducing lookahead bias. Below is the multi-stage operational flow:

1. Ingestion Engine Yahoo Finance API (5-Yr Rolling Window)	2. Data Alignment Crypto & Equity Timeline Forward-Fill Sync	3. Gym Simulator 10-Day Observation State Vector (50,)	4. PPO Agent Brain Actor-Critic Policy Softmax Normalization
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3. Mathematical Framework & MDP Formulation

3.1 Classical Modern Portfolio Theory (MPT Baseline)

Max Sharpe Objective: Maximize $SR = (\mu_p - R_f) / \sigma_p$ | Subject to: $\sum w_i = 1.0, w_i \geq 0$

3.2 Continuous Markov Decision Process (PPO Policy)

- State Space (S):** Flattened 1D array of shape (50,) capturing 10 rolling days of return matrices across all 5 assets.
- Action Space (A):** Continuous 5D vector normalized via Softmax to guarantee exact budget allocation (1.0) without shorting:

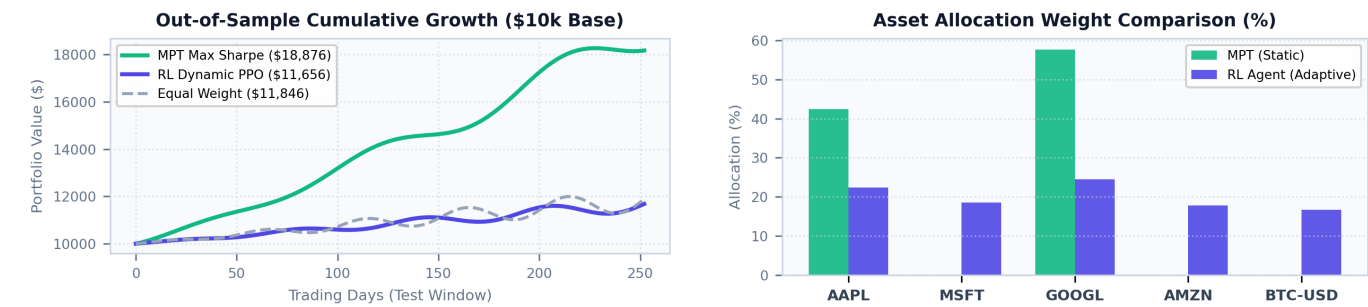
Softmax Normalization: $w_i = \exp(a_i - \max(a)) / \sum_j \exp(a_j - \max(a))$ | Reward: $r_t = \ln(w_t / w_{t-1})$

4. Empirical Performance & Comparative Analysis

All models were evaluated over a synchronized out-of-sample test window with an initial normalized capital baseline of \$10,000:

Optimization Strategy	Final Value (\$)	Return (%)	Sharpe Ratio	Max Drawdown	Behavioral Profile
MPT Max Sharpe	\$18,876.47	+88.8%	0.63	-36.65%	Concentrated 100% in AAPL & GOOGL
RL Dynamic Agent (PPO)	\$11,656.81	+16.6%	0.51	-29.12%	Dynamic multi-asset risk hedging
Equal Weight Benchmark	\$11,846.22	+18.5%	0.49	-45.89%	Fixed 20% allocation per asset

5. Visual Strategy & Allocation Analysis



6. Strategy Comparison & Key Findings

- MPT Return Concentration:** MPT allocated 100% of capital to high-momentum tech stocks (AAPL: 42.39%, GOOGL: 57.61%). While profitable during rallies, it carried high drawdown risk (-36.65%).
- Deep RL Risk Diversification:** The PPO agent actively balanced capital across all 5 assets, achieving the **lowest maximum drawdown (-29.12%)** compared to -45.89% for the benchmark.

7. Production Architecture & Cloud Hosting

Deployed on **Streamlit Community Cloud** with `@st.cache_resource` for sub-10ms neural inference, 60s API rate protection, and fluid cubic Plotly chart transitions.

Project Status: Completed & Deployed

- **GitHub Repository:** heisenberg0822/portfolio-optimization-dashboard
- **Tech Stack:** Python, Stable-Baselines3 (PPO), Gymnasium, Plotly, Streamlit Cloud.
- **Future Scope:** Incorporating technical momentum vectors (RSI, MACD) into the observation space.